

Traffic Light Controller Using Verilog HDL

Finite State Machine Based RTL Design, Simulation and Verification

By

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1 Project Overview

Traffic management is a critical component of modern transportation systems, ensuring the safe and efficient movement of vehicles through road intersections. Traditional traffic signal systems often rely on fixed timing schedules to regulate vehicle flow and minimize congestion. A Traffic Light Controller (TLC) is a digital control system that manages traffic signals by coordinating the sequence and duration of red, yellow, and green lights.

This project presents the design and verification of a Traffic Light Controller using Verilog HDL. The controller is implemented as a Finite State Machine (FSM) with four operational states representing different traffic signal configurations for the main road and small road. A counter-based timing mechanism is employed to control the duration of each state and facilitate state transitions.

The design was developed using RTL design methodology and verified through simulation using Vivado. The functionality of the controller was validated by observing correct state transitions, timing behavior, and signal outputs under various operating conditions. The project demonstrates the application of FSM-based digital design techniques in real-world traffic management systems and provides a foundation for developing more advanced intelligent traffic control solutions.

2 System Requirements and Specifications

This Traffic Light Controller (TLC) is designed to regulate traffic flow between a main road and a small road using a Finite State Machine (FSM) based control mechanism. The controller generates appropriate traffic signal outputs based on predefined timing intervals and state transitions.

2.1 Functional Requirements

The Traffic Light Controller shall satisfy the following functional requirements:

- Control traffic signals for both the main road and small road.
- Generate Green, Yellow, and Red light indications.
- Operate using a finite state machine with four distinct states.
- Transition between states based on predefined timing intervals.
- Reset to a known initial state upon activation of the reset signal.
- Continuously cycle through the traffic signal sequence.

2.2 Input and Output Interface

The module interfaces used in the design are listed below.

Table 1: Input and Output Signals

Signal	Direction	Description
clk	Input	System Clock Signal
rst	Input	Asynchronous Reset Signal
SR[2:0]	Output	Small Road Traffic Signals
MR[2:0]	Output	Main Road Traffic Signals
counter[7:0]	Output	State Timing Counter

2.3 Timing Specifications

The controller utilizes an 8-bit counter to determine the duration of each traffic signal state. The counter increments on every rising edge of the clock and resets after reaching a count value of 75. State transitions occur when the counter reaches specific threshold values, thereby defining the active duration of each traffic signal state.

The timing allocation for each state is summarized in the Timing Sequence Analysis section of this report.

3 System Architecture

The Traffic Light Controller is implemented using a Finite State Machine (FSM) consisting of four operational states. Each state corresponds to a unique traffic signal configuration for the main road and small road. The controller transitions between these states based on predefined timing intervals generated using an 8-bit counter.

The overall architecture consists of the following major components:

- State Register
- State Transition Logic
- Timing Counter
- Output Generation Logic
- Reset Logic

Together, these components ensure proper sequencing of traffic signals while maintaining safe traffic flow between the two roads.

3.1 Traffic Signal Configuration

The controller manages traffic signals for two roads:

- Main Road (MR)
- Small Road (SR)

Each road contains three traffic signal indicators:

- Green
- Yellow
- Red

At any given time, only one signal is active for a particular road. The signal configuration is determined by the current FSM state.

3.2 State Encoding

Each FSM state is represented using a two-bit binary code. The state encoding scheme simplifies the implementation of state transitions and output generation logic.

TABLE 1: STATE ENCODING

State	Binary Encoding
G_R (Small Road Green, Main Road Red)	2'b00
G_Y (Small Road Green, Main Road Yellow)	2'b01
R_G (Small Road Red, Main Road Green)	2'b10
Y_G (Small Road Yellow, Main Road Green)	2'b11

3.3 State Transition Diagram

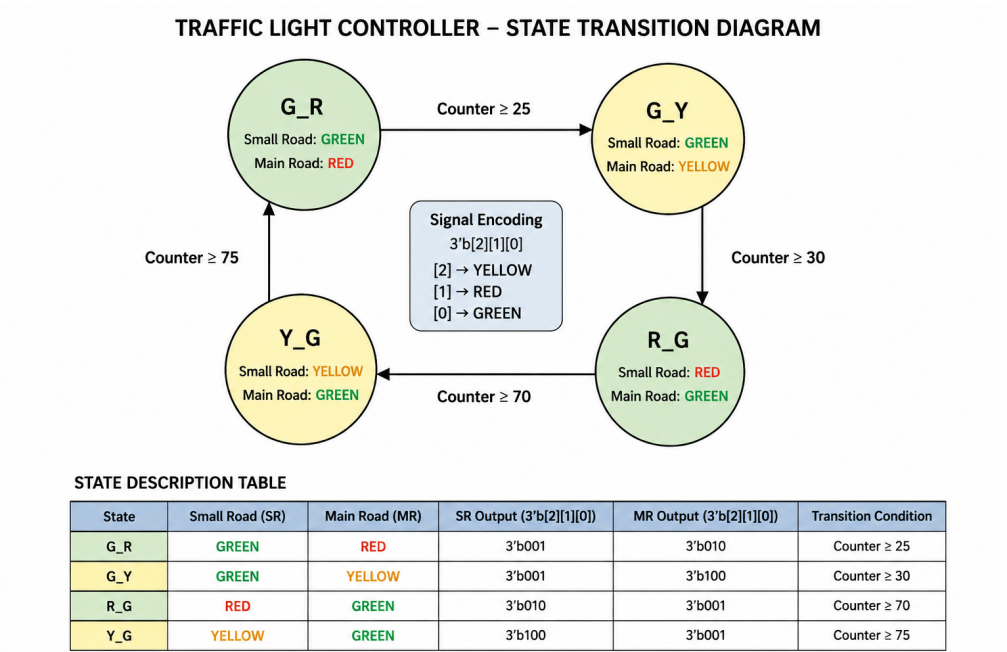


Figure 1: Traffic Light Controller State Transition Diagram

The state transition diagram illustrates the movement between FSM states based on counter threshold values. The controller continuously cycles through the four operational states to regulate traffic flow.

3.4 Timing Sequence Analysis

The duration of each traffic signal state is controlled using an 8-bit counter. State transitions occur when the counter reaches predefined threshold values. This timing mechanism ensures predictable and repeatable traffic signal operation.

TABLE 2: TIMING SEQUENCE

State	Counter Range	Duration (Clock Cycles)
G_R (SR Green, MR Red)	1 – 25	25 Cycles
G_Y (SR Green, MR Yellow)	26 – 30	5 Cycles
R_G (SR Red, MR Green)	31 – 70	40 Cycles
Y_G (SR Yellow, MR Green)	71 – 75	5 Cycles

Figure 2: Traffic Signal Timing Allocation

4 RTL Design and Implementation

4.1 Complete RTL Code

```

module TLC(input clk,rst ,  output reg [2:0]SR,MR, output reg [7:0]counter);
    reg [1:0] st;
    parameter G_R = 2'b00, G_Y = 2'b01, R_G = 2'b10, Y_G = 2'b11;

    always @(posedge clk or posedge rst) begin
        if(rst)
            counter <= 1;
        else begin
            if(counter == 75)
                counter <= 1;
            else counter <= counter + 1;
        end
    end

    always @(posedge clk or posedge rst)begin
        if(rst)
            st <= G_R;
        else begin

```

```

        case(st)
            G_R : st <= (counter >= 25) ? G_Y : G_R ;
            G_Y : st <= (counter >= 30) ? R_G : G_Y ;
            R_G : st <= (counter >= 70) ? Y_G : R_G ;
            Y_G : st <= (counter >= 75) ? G_R : Y_G ;
            default : st <= G_R;
        endcase
    end
end
always @ * begin
    case(st)
        G_R : begin SR = 3'b001; MR = 3'b010; end
        G_Y : begin SR = 3'b001; MR = 3'b100; end
        R_G : begin SR = 3'b010; MR = 3'b001; end
        Y_G : begin SR = 3'b100; MR = 3'b001; end
    endcase
end
endmodule

```

4.2 Design Methodology

The Traffic Light Controller was designed using a Finite State Machine (FSM) architecture. Four states were defined to represent different traffic signal combinations for the main road and small road. A counter-based timing mechanism was employed to control the duration of each state and trigger state transitions.

The design consists of three primary functional blocks:

- Counter Logic
- State Transition Logic
- Output Generation Logic

The counter continuously increments on every rising edge of the clock and determines when the FSM should transition to the next state. Based on the current state and counter value, the controller updates the traffic signal outputs accordingly.

4.3 Counter-Based Timing Logic

The timing functionality of the controller is implemented using an 8-bit counter. The counter increments on every positive edge of the clock signal and resets to a value of one whenever the reset signal is asserted or when the count reaches seventy-five.

This counter acts as the timing reference for the entire controller. Different counter threshold values are used to determine the duration of each traffic signal state. By comparing the counter value against predefined limits, the controller transitions between states at the required time intervals.

The use of a dedicated counter simplifies timing control and allows predictable traffic signal operation throughout the FSM cycle.

4.4 State Transition Logic

The controller utilizes a four-state FSM to manage traffic signal sequencing. State transitions are determined by the current state and the counter value.

The FSM begins operation in the G_R state after reset. When the counter reaches the specified threshold value, the controller transitions to the next state in the sequence. The state progression follows the order:

$$G_R \rightarrow G_Y \rightarrow R_G \rightarrow Y_G \rightarrow G_R$$

This cyclic transition pattern ensures continuous and orderly traffic signal control for both roads.

4.5 Output Generation Logic

The output generation logic determines the traffic signal indications for the main road and small road based on the current FSM state.

For each state, specific output combinations are assigned to the signal vectors SR and MR. These vectors control the Green, Red, and Yellow indicators for the respective roads.

The output logic is implemented using combinational logic, ensuring that the traffic signals are updated immediately whenever the FSM enters a new state. This approach provides clear separation between state transition logic and output generation logic, improving design readability and maintainability.

5 Verification Environment

5.1 Complete Testbench Code

The functionality of the Traffic Light Controller was verified using a dedicated Verilog testbench. The complete testbench used for simulation and verification is shown below.

```
module TLC_tb;
    reg clk, rst;
    wire [2:0] SR, MR;
    wire [7:0] counter;

    TLC dut(clk, rst, SR, MR, counter);
    always #5 clk = ~clk;

    initial begin
        $monitor("Time = %0d -> clk = %b, rst = %b, state = %b, counter = %d",
            $time, clk, rst, dut.st, counter );
    end
endmodule
```



```

    clk = 0; rst = 1; #10;
    rst = 0; #800;
    $stop;
end
endmodule

```

6 Results and Analysis

6.1 RTL Schematic Analysis

The RTL schematic generated by Vivado provides a hardware-level view of the Traffic Light Controller. It shows how the different parts of the design are connected together to implement the required traffic signal operation.

The schematic includes the state register, counter, state transition logic, and output logic. The counter is used to control the duration of each traffic signal state, while the FSM determines the current operating state of the controller. Based on the active state, the appropriate traffic signal outputs are generated for the Main Road (MR) and Small Road (SR).

The RTL schematic confirms that the Verilog design has been synthesized successfully and that all functional blocks are interconnected correctly. This verifies that the controller can be implemented using digital hardware resources.

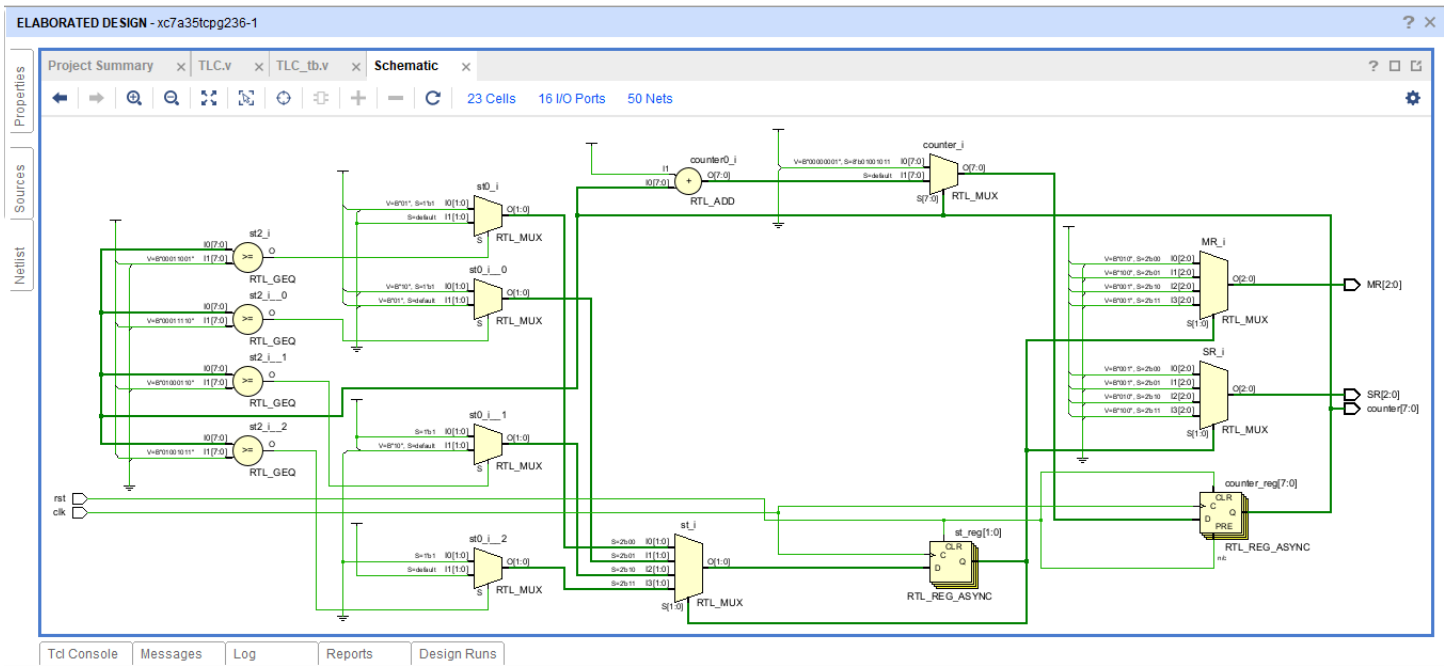


Figure 3: RTL Schematic of the Traffic Light Controller

6.2 Behavioral Simulation Results

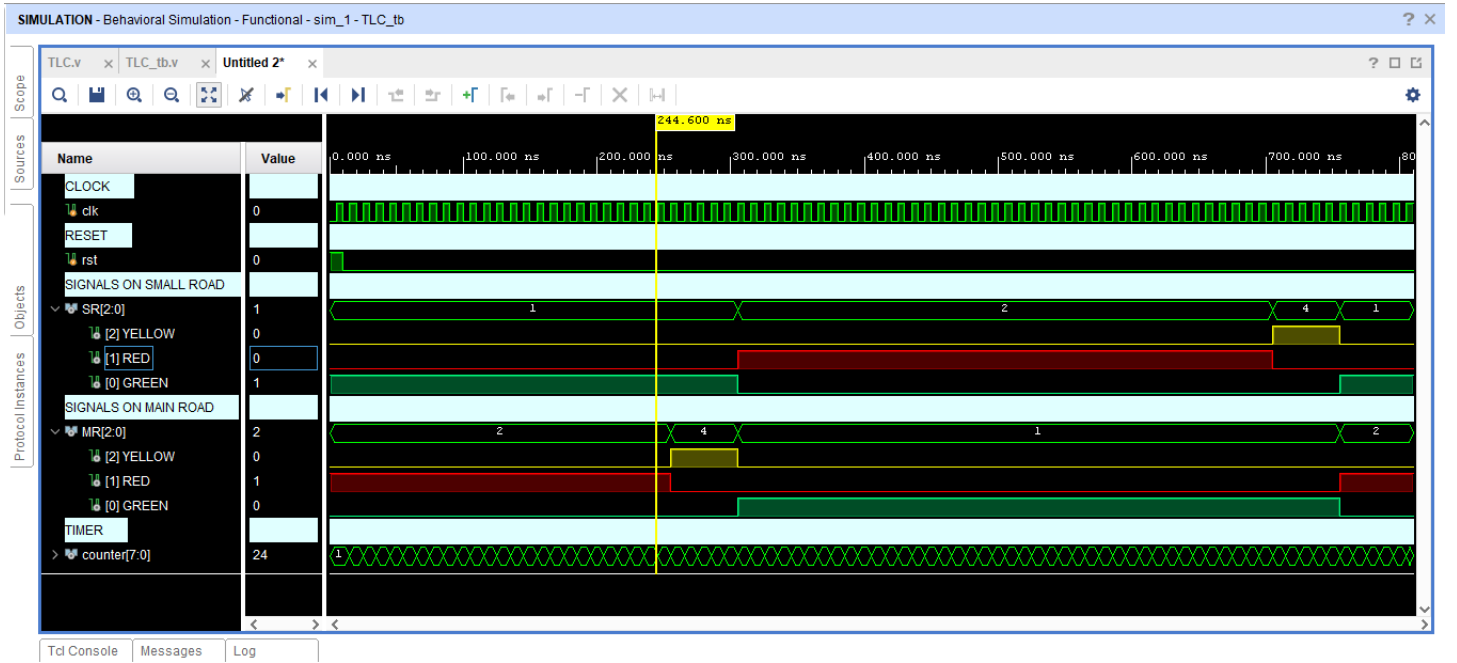


Figure 4: Behavioral Simulation of the Traffic Light Controller

The behavioral simulation was performed to verify the functional correctness of the Traffic Light Controller. The waveform confirms that the FSM transitions through the states in the intended sequence and that the traffic signal outputs change according to the active state.

The counter increments continuously and initiates state transitions when predefined threshold values are reached. The observed state progression matches the design specification:

$$G_R \rightarrow G_Y \rightarrow R_G \rightarrow Y_G \rightarrow G_R$$

The simulation results demonstrate stable and predictable operation of the controller throughout the complete traffic signal cycle.

6.3 Timing Verification

Timing verification was performed by observing the relationship between the counter values and the corresponding state transitions. The simulation results confirm that each traffic signal state remains active for the intended duration before transitioning to the next state.

The counter increments on every rising edge of the clock and resets after reaching a value of 75. The observed timing behavior matches the design specifications, ensuring predictable and reliable operation of the controller.

The successful timing verification demonstrates the correctness of the counter-based timing mechanism used in the Traffic Light Controller.

7 Design Insights

7.1 Challenges Encountered

During the development of the Traffic Light Controller, one of the primary challenges was determining the appropriate timing intervals for each traffic signal state and implementing them correctly using the counter. Careful attention was required to ensure that state transitions occurred at the intended counter values.

Another challenge was verifying the correctness of the FSM behavior during simulation. Monitoring the internal state register and counter values helped in validating the state transition sequence and identifying any unexpected behavior during the design process.

7.2 Key Learnings

This project provided practical experience in designing Finite State Machines (FSMs) using Verilog HDL. It improved understanding of state encoding, state transition logic, and output generation techniques.

The project also reinforced the importance of simulation and verification in digital design. Through waveform analysis and RTL schematic inspection, the functionality of the controller was validated before hardware implementation.

Overall, the project strengthened skills in RTL design, debugging, simulation, and FSM-based digital system development.

8 Future Enhancements

8.1 Vehicle Density-Based Traffic Control

The current design uses fixed timing intervals for all traffic conditions. Future versions can incorporate vehicle density sensors to dynamically adjust signal durations based on traffic flow.

8.2 Pedestrian Crossing Support

Pedestrian crossing signals can be integrated into the controller to improve road safety. Dedicated pedestrian states can be added to the FSM for controlled crossing operations.

8.3 Emergency Vehicle Priority

The controller can be enhanced to detect emergency vehicles such as ambulances and fire trucks. This feature would allow priority signal switching to facilitate faster emergency response.

8.4 Dynamic Timing Adjustment

Traffic signal durations can be automatically modified based on real-time road conditions. This would improve traffic efficiency during peak and non-peak hours.

8.5 FPGA Hardware Implementation

The design can be deployed on an FPGA platform for real-time testing and validation. Hardware implementation would provide practical insight into timing and resource utilization.

8.6 Multi-Road Intersection Support

The current controller manages a single intersection with two roads. Future versions can be extended to support complex multi-road intersections with additional traffic signals.

9 Conclusion

This project successfully demonstrated the design and verification of a Traffic Light Controller (TLC) using Verilog HDL. The controller was implemented as a four-state Finite State Machine (FSM) that regulates traffic signals for a main road and a small road using a counter-based timing mechanism.

The RTL design was developed and verified through simulation in Vivado. The simulation results confirmed correct state transitions, timing behavior, and traffic signal generation according to the design specifications. The generated RTL schematic further validated the successful synthesis of the design.

Through this project, practical experience was gained in FSM design, RTL development, simulation, debugging, and verification methodologies. The Traffic Light Controller serves as a fundamental example of how digital design techniques can be applied to real-world control systems and provides a strong foundation for developing more advanced traffic management solutions.